

merrypopins: A Python package for nanoindentation data science

Cahit Acar¹, Anna Marcelissen¹, Hugo van Schroyen Lantman¹, and John M. Aiken^{1,2,3}✉

¹ Utrecht University, The Netherlands ² Expert Analytics, Norway ³ University of Oslo, Njord Centre, Norway ✉ Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) ✉
- [Repository](#) ✉
- [Archive](#) ✉

Editor: Fei Tao ✉

Reviewers:

- [@KennethKwabenaKenney](#)
- [@Manishms18](#)

Submitted: 22 July 2025

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](https://creativecommons.org/licenses/by/4.0/)).

Summary

merrypopins is a Python library that streamlines nanoindentation data processing, automated pop-in detection, and statistical analysis of collections of pop-in events. Nanoindentation deforms materials experimentally to characterise their behaviour and quantify mechanical properties from load-displacement data (Oliver & Pharr, 1992, 2004). Experiments performed with a spherical tip can also be used to construct stress-strain curves and, by extension, determine the yield point, defining the transition from elastic to plastic deformation (Kalidindi & Pathak, 2008; Pathak & Kalidindi, 2015). Understanding the onset of plasticity at the microscale matters for applications in engineered materials and earthquake mechanics. A common feature during the loading part of these experiments is a sudden increase of indentation depth at constant force, called a “pop-in”. Recognising pop-ins by hand is labour-intensive and subjective, which makes automated, reproducible detection important.

Statement of need

Pop-ins are difficult to detect because they appear in subtle, intermittent, and varied ways within indentation curves. Analysts have historically picked them by hand from depth vs. time or stress-strain curves, an approach that is subjective, labour-intensive, and inconsistent between observers. Modern nanoindentation machines can perform up to 12 indentations per second (Bruker, n.d.), so the volume of data has outgrown what manual inspection can keep up with.

Pop-ins are linked to dislocation motion in crystalline materials and are considered small-scale analogues of earthquakes (Ispánovity et al., 2022; Sato et al., 2020), following the same kind of statistical patterns, such as power-law distributions in size and in waiting time between events. The first pop-in generally coincides with the onset of plasticity, and the stress at which it occurs depends strongly on indenter tip size and on pre-existing plastic deformation (Morris et al., 2011; Shim et al., 2008), and hence on the apparent yield hardness. A small tip stresses a volume more likely to be dislocation-free, so the material stays elastic up to higher stress, whereas a larger tip samples more volume and yields earlier. Both correcting for this size effect and recovering the underlying statistics require every event along a curve rather than only the first, which is precisely what manual picking scales worst at.

The primary users of merrypopins are students and researchers in materials science, geology, nano-mechanics, and earthquake science, fields in which high-resolution indentation is increasingly used to investigate plastic and fracture processes at the microscale. Despite a growing number of studies targeting pop-ins in load-depth curves, almost all rely on manual inspection or on private scripts whose methods are never disclosed, leaving no accessible, reproducible event detection software to build on. What is needed is an adaptable, open-source tool that works out of the box for non-programmers and stays extensible for power users.

42 New detection techniques, parameter settings, and visualisation modules can be contributed
43 through the public GitHub repository.

44 State of the field

45 Software for instrumented indentation falls into three groups.

46 First, vendor software shipped with commercial indenters, such as the suite supplied with the
47 Bruker Hysitron platform (Bruker, n.d.), acquires curves and computes hardness and modulus.
48 It is closed-source, tied to one instrument, and offers no pop-in extraction, so analyses built on
49 it cannot be reproduced outside the vendor's environment.

50 Second, open-source indentation packages target property extraction rather than event detection.
51 micromechanics (Brinckmann & Micromechanics Team, 2026) reads a wide range of vendor
52 formats and evaluates hardness and Young's modulus by the Oliver-Pharr method (Oliver &
53 Pharr, 1992, 2004), but has no pop-in detection layer.

54 Third, a few tools address pop-ins directly. The closest is the PopIn toolbox (Mercier, 2023), a
55 MATLAB package that fits the Hertz model to load-displacement curves and plots Weibull or
56 time- and temperature-dependent cumulative distributions of the first or second pop-in. Two
57 things make it a poor foundation for our target audience: it requires a commercial MATLAB
58 licence, and it is scoped to the first and second pop-in of a curve rather than to every event
59 along it. Power-law size and waiting-time statistics need the full catalogue. Machine learning
60 has been applied as well: Kossman & Bigerelle (2021) trained a convolutional neural network
61 that classifies whole curves as containing a pop-in or not, at roughly 93% accuracy. That
62 answers a different question: whether a curve contains a pop-in, not where each one sits. It
63 also needs labelled training curves, and the code was not released.

64 merrypopins therefore fills a gap. Its contribution is fourfold: per-point localisation of every
65 pop-in along a curve instead of per-curve classification; four complementary detectors behind
66 one interface, with an explicit cross-method agreement score; a statistical suite that turns
67 located events into the stress-strain, precursor, and temporal quantities that pop-in studies
68 report; and a no-code Streamlit app for users who do not program. All of it is MIT-licensed
69 Python installable from PyPI.

70 Software design

71 merrypopins is five modules that pass pandas DataFrames along a pipeline: load_datasets
72 parses vendor .txt curves and .tdm/.tdx metadata, preprocess trims setup artefacts and
73 zeroes the depth axis at the detected contact point, locate detects pop-ins, statistics
74 computes stress-strain curves, yield points, and precursor and temporal statistics, and
75 make_dataset runs the first three end-to-end and writes annotated tables with overlay plots.
76 Because every stage consumes and returns a DataFrame, users can stop anywhere, inspect
77 the result, or substitute their own step. This was a deliberate trade-off: a single opaque
78 analyse() call would be simpler to document, but researchers need to see and correct the
79 contact-point detection before trusting anything downstream.

80 **Choosing and combining detection methods.** Four detectors suit different material systems and
81 experimental circumstances. Savitzky-Golay smoothing suppresses high-frequency noise while
82 preserving curve shape (Savitzky & Golay, 1964), and Fourier spectral differentiation identifies
83 abrupt discontinuities with minimal parameterisation (Cooley et al., 1969); both are cheap and
84 interpretable, a good first pass over a large batch. Two unsupervised methods adapt to the
85 data: an Isolation Forest finds anomalies in a multidimensional feature space without labelled
86 instances (Liu et al., 2008), useful when pop-in magnitudes are unknown beforehand, and a
87 convolutional autoencoder learns representations directly from the curve, catching nonlinear
88 patterns a fixed derivative threshold overlooks (Malhotra et al., 2016), at higher cost. Rather

89 than asking the user to pick one, `default_locate` runs the enabled methods and reports all
90 of them: each writes its own boolean column, `popin` is their union, `popin_score` counts how
91 many methods fired at a point, and `popin_confident` requires agreement from at least two.
92 Users who need recall take `popin`; users for whom a false positive costs more than a missed
93 event take `popin_confident`.

94 **No pretrained models.** Neither learning method ships with weights. Both are fitted,
95 unsupervised, on the curve being analysed at call time, using local stiffness difference and
96 curvature as features, so no labelled training data is needed and a result depends only on
97 the curve and parameters supplied. All four methods are restricted to the loading portion of
98 the curve, up to maximum load, since unloading artefacts are irrelevant to pop-in analysis.
99 TensorFlow, needed only by the autoencoder, is an optional install extra so the other three
100 carry no heavyweight dependency.

101 **Validating detections.** Because pop-in ground truth is itself expert judgement, detections are
102 built to be checked rather than merely trusted: `popin_score` makes cross-method agreement
103 visible, flagging events only one detector saw, and `make_dataset` writes an overlay plot per
104 curve with each method's detections in a distinct colour, so a whole batch can be reviewed
105 visually.

106 **Development approach.** The library was built with a tutorial-driven development framework
107 (Aiken, 2020; Aiken et al., 2025; Woods et al., 2022). Rather than starting from architectural
108 specifications, the scientist's process is written as an executable lesson, typically a Jupyter
109 notebook: researchers write function stubs inside a scientific narrative and developers implement
110 them against real usage cases. Scientific usability drives the design.

111 Research impact statement

112 `merrypopins` is released for use rather than as a prototype: MIT licensed, installable from
113 PyPI, documented, continuously tested in CI, thirteen tagged releases archived on Zenodo
114 (10.5281/zenodo.21906346), with contributing guidelines and a public issue tracker.

115 Its near-term significance rests on being usable and checkable by its target community. The
116 repository ships 36 real indentation curves with instrument metadata and rendered overlay
117 plots, five tutorial notebooks reproducing the single-file and multi-file workflows, and a test
118 suite covering all five modules, including synthetic curves with pop-ins of known position and
119 size. Detector behaviour and every documented workflow can therefore be re-run independently.
120 The hosted Streamlit app puts the same pipeline in front of experimentalists who do not
121 program, removing the installation barrier that keeps analysis code out of many indentation
122 laboratories.

123 It was developed on data from the Utrecht University indentation laboratory to support
124 nanoindentation work in the SerpRateAI project and EPOS-eNLarge, and is the analysis layer
125 for that ongoing research.

126 Code availability

127 The `merrypopins` package can be installed via:

```
128 pip install merrypopins          # three detectors, lightweight  
129 pip install 'merrypopins[cnn]'   # adds the autoencoder, pulls in TensorFlow
```

130 The source is on GitHub (<https://github.com/SerpRateAI/merrypopins>), together with the
131 datasets, tutorial notebooks, and test suite described above; contributions are welcome by
132 forking the repository and opening a pull request. The Streamlit app is at [https://merrypopins.
133 streamlit.app/](https://merrypopins.streamlit.app/).

134 AI usage disclosure

135 The merrypopins library, its test suite, its documentation, and the original version of this
136 manuscript were written by the authors without generative AI assistance.

137 During revision in response to peer review, an AI coding assistant was used for three things:
138 drafting prose for the “State of the field”, “Software design”, and “Research impact statement”
139 sections from sources the authors selected and checked, adding type annotations to existing
140 function signatures, and refactoring TensorFlow into an optional install extra. Every resulting
141 change was reviewed, tested, and accepted by the authors. All scientific claims, design
142 decisions, and comparisons with other software are the authors’ own, and the authors take full
143 responsibility for both the software and this paper.

144 Acknowledgements

145 This project has received funding from the Norwegian Research Council (SerpRateAI, grant no.
146 334395) and is supported by EPOS-eNLarge funded by the Dutch Research Council (NWO)
147 Roadmap for large-scale research infrastructure. We would like to thank Alissa Kotowski for
148 fruitful conversations.

149 References

- 150 Aiken, J. M. (2020). *Science coding vs software development*. Substack. <https://mnky9800n.substack.com/p/science-coding-vs-software-development>
- 151
- 152 Aiken, J. M., Jones, G., Yin, X., Abele, A. K., Woods, C., Westaway, R. M., & Bamber, J.
153 L. (2025). 4DModeller: A spatio-temporal modelling package. *Journal of Open Source*
154 *Software*, 10(106), 7047. <https://doi.org/10.21105/joss.07047>
- 155 Brinckmann, S., & Micromechanics Team. (2026). *Micromechanics: Investigation*
156 *of nanoindentation and SEM data*. GitHub. <https://micromechanics.github.io/micromechanics/>
- 157
- 158 Bruker. (n.d.). *Hysitron TI 990 TriboIndenter brochure*. Bruker. <https://www.bruker.com/en/meta/forms/bns-form-pages/brochures/ni/hysitron-ti-990.html>
- 159
- 160 Cooley, J. W., Lewis, P. A. W., & Welch, P. D. (1969). The fast Fourier transform and its
161 applications. *IEEE Transactions on Education*, 12(1), 27–34. <https://doi.org/10.1109/TE.1969.4320436>
- 162
- 163 Ispánovity, P. D., Ugi, D., Péterffy, G., Knappek, M., Kalácska, S., Tüzes, D., Dankházi, Z.,
164 Máthjs, K., Chmelík, F., & Groma, I. (2022). Dislocation avalanches are like earthquakes on
165 the micron scale. *Nature Communications*, 13(1), 1975. <https://doi.org/10.1038/s41467-022-29044-7>
- 166
- 167 Kalidindi, S. R., & Pathak, S. (2008). Determination of the effective zero-point and the
168 extraction of spherical nanoindentation stress-strain curves. *Acta Materialia*, 56(14),
169 3523–3532. <https://doi.org/10.1016/j.actamat.2008.03.036>
- 170 Kossman, S., & Bigerelle, M. (2021). Pop-in identification in nanoindentation curves with
171 deep learning algorithms. *Materials*, 14(22), 7027. <https://doi.org/10.3390/ma14227027>
- 172 Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation forest. *2008 Eighth IEEE International*
173 *Conference on Data Mining*, 413–422. <https://doi.org/10.1109/ICDM.2008.17>
- 174 Malhotra, P., Ramakrishnan, A., Anand, G., Vig, L., Agarwal, P., & Shroff, G. (2016). *LSTM-*
175 *based encoder-decoder for multi-sensor anomaly detection*. <https://doi.org/10.48550/arXiv.1607.00148>
- 176

- 177 Mercier, D. (2023). *PopIn: "Pop-in" analysis from indentation tests*. GitHub. <https://github.com/DavidMercier/PopIn>
- 178
- 179 Morris, J. R., Bei, H., Pharr, G. M., & George, E. P. (2011). Size effects and stochastic
180 behavior of nanoindentation pop in. *Physical Review Letters*, 106(1), 165502. <https://doi.org/10.1103/PhysRevLett.106.165502>
- 181
- 182 Oliver, W. C., & Pharr, G. M. (1992). An improved technique for determining hardness and
183 elastic modulus using load and displacement sensing indentation experiments. *Journal of*
184 *Materials Research*, 7(6), 1564–1583. <https://doi.org/10.1557/JMR.1992.1564>
- 185 Oliver, W. C., & Pharr, G. M. (2004). Measurement of hardness and elastic modulus by
186 instrumented indentation: Advances in understanding and refinements to methodology.
187 *Journal of Materials Research*, 19(1), 3–20. <https://doi.org/10.1557/jmr.2004.19.1.3>
- 188 Pathak, S., & Kalidindi, S. R. (2015). Spherical nanoindentation stress-strain curves. *Materials*
189 *Science and Engineering R*, 91(1), 1–36. <https://doi.org/10.1016/j.mser.2015.02.001>
- 190 Sato, Y., Shinzato, S., Ohmura, T., Hatano, T., & Ogata, S. (2020). Unique universal scaling
191 in nanoindentation pop-ins. *Nature Communications*, 11(1), 4177. <https://doi.org/10.1038/s41467-020-17918-7>
- 192
- 193 Savitzky, A., & Golay, M. J. E. (1964). Smoothing and differentiation of data by simplified
194 least squares procedures. *Analytical Chemistry*, 36(8), 1627–1639. <https://doi.org/10.1021/ac60214a047>
- 195
- 196 Shim, S., Bei, H., George, E. P., & Pharr, G. M. (2008). A different type of indentation size
197 effect. *Scripta Materialia*, 59(1), 1095–1098. <https://doi.org/10.1016/j.scriptamat.2008.07.026>
- 198
- 199 Woods, C., Hedges, L., Edsall, C., Brooks-Pollock, E., Parton-Fenton, C., McKinley, T. J.,
200 Keeling, M. J., & Danon, L. (2022). MetaWards: A flexible metapopulation framework
201 for modelling disease spread. *Journal of Open Source Software*, 7(70), 3914. <https://doi.org/10.21105/joss.03914>
- 202